



Semiconductor Manufacturing International Corporation

Quick-start on SMIC Calibre LVS - xRC Flow

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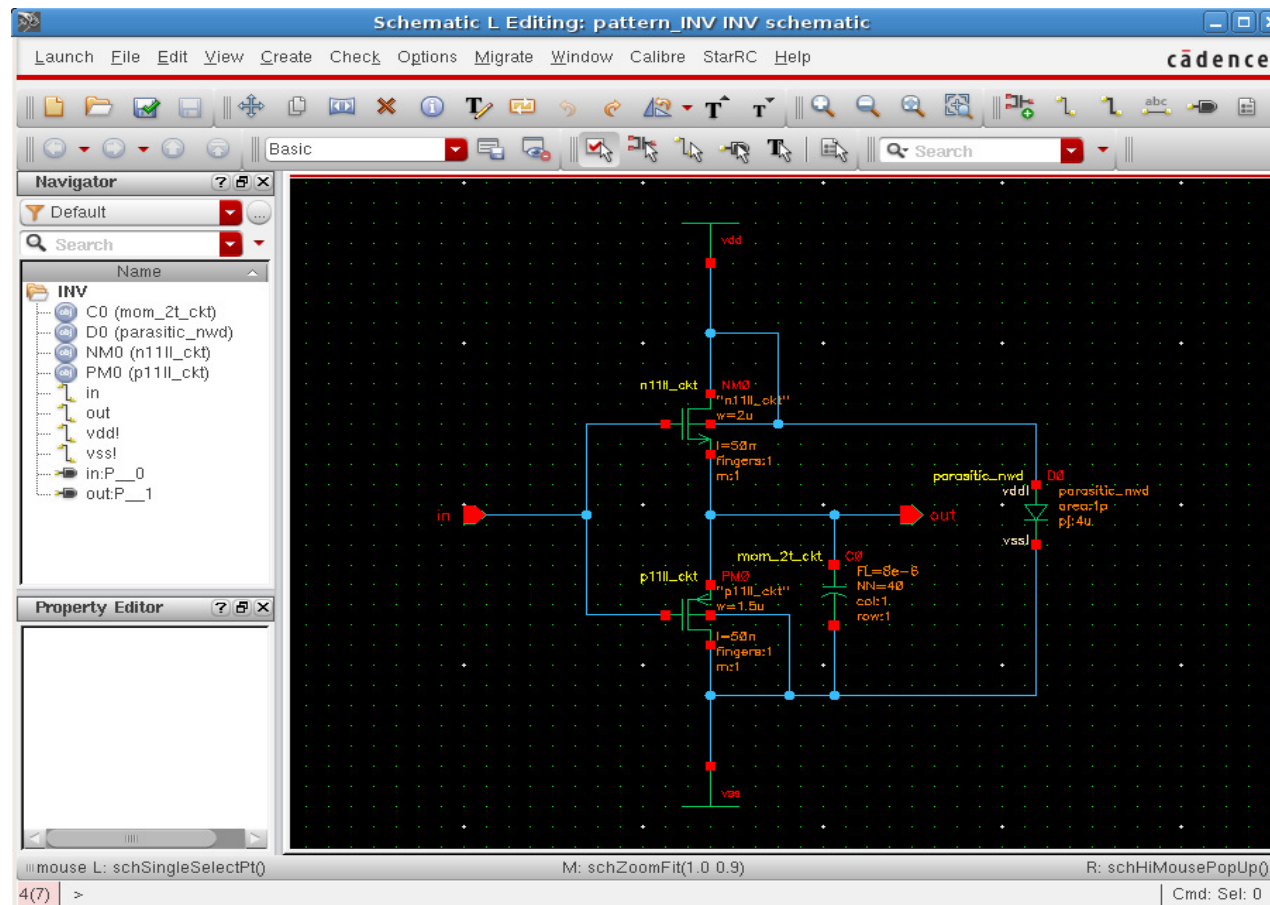
Before You Start to Run Calibre xRC

- SMIC recommends user to execute parasitic extraction based on a LVS-clean database. Therefore, you need the following components when you execute Calibre xRC,
 - The CDL / SPICE netlist of your design
 - The corresponding layout GDS of your design
 - *<Calibre_LVS_runset>*, you could get from SMIC-Now website
 - Calibre xRC runsets (include *hcell_list* and *xcell_list*), you could get from SMIC-Now website
- There are two ways to run Calibre xRC: batch mode and GUI mode. Both of the working mode will be demonstrated here.
- This tutorial is a general example of Calibre LVS - xRC flow. For advanced features and application of Calibre xRC, please consult Mentor Graphics (<http://www.mentor.com/>) for more information.

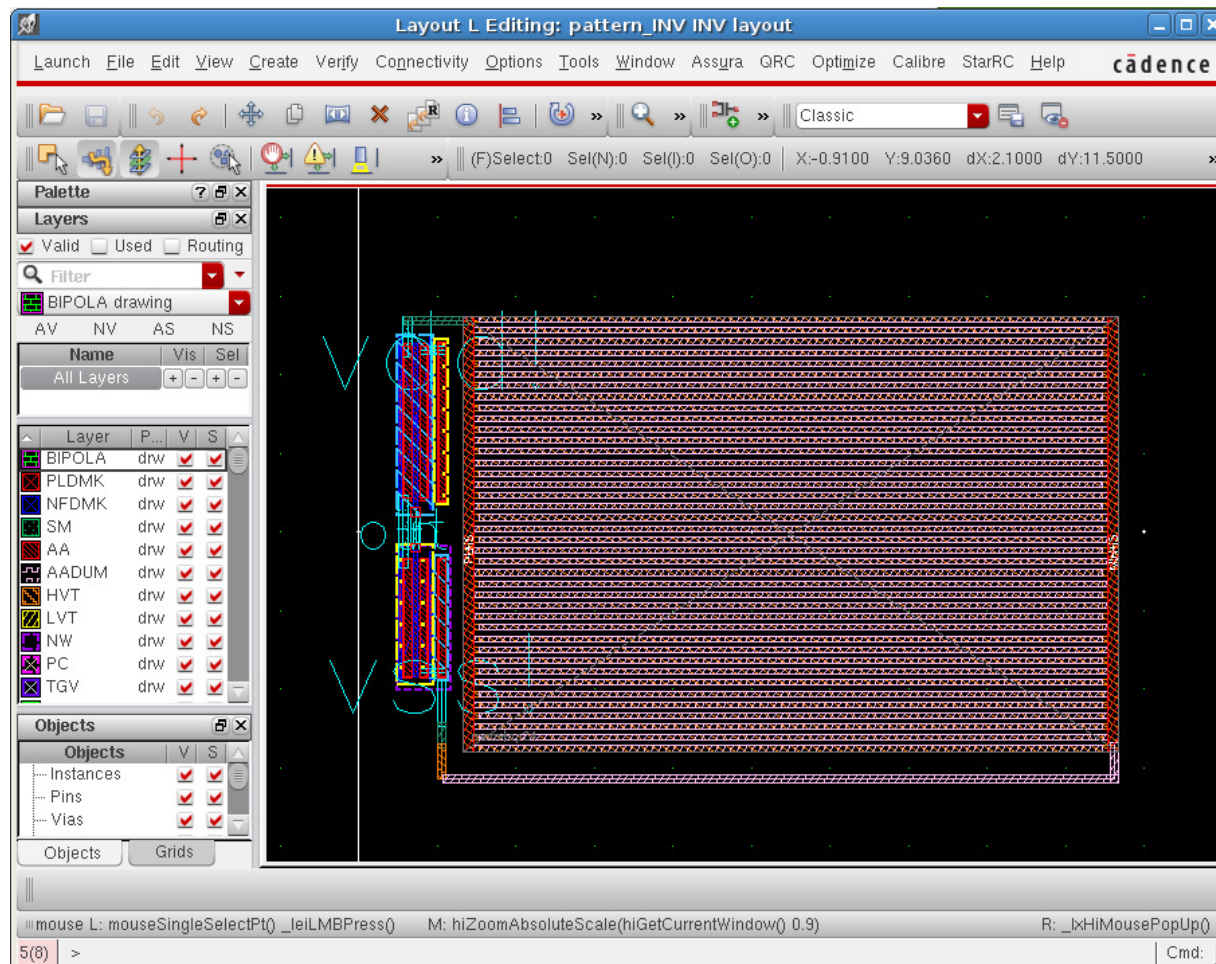


Schematic of demonstration

- Here we use SMIC 40LLRF PDK to design an Inverter for demonstration.
- Capacitance mom_2t_ckt device should be blocked in RC extraction.



Schematic-Driven-Layout





[Batch mode] STEP 1: Complete Setups in LVS Runset (Conti.)

- Specify the *Environment Setting* section in Calibre LVS runset, based on SMIC process and metallization options you preferred, e.g.
 - **TOPMETAL** define the total metal layers you preferred
 - **TOP_METAL_NUM** define the top metal layers you preferred
- Set **XRC_RUN TRUE**

```
//=====
/**OPTION 2: Define total number of Metal.The value can be 10~4. Default value is 10.
#DEFINE TOPMETAL 9
//=====
/**OPTION 3: Define type of top metal.The value can be single or double.
/**Default value is double.
/**If you use double top metal process, set this option to double(Lower Case).
/**If you use single top metal process, set this option to single(Lower Case).
/**If you set TOPMETAL to 10, you must set option TOP_METAL_NUM to double, single top
/**metal for 1P10M is not allowd!
/**If you set TOPMETAL to 4, you must set option TOP_METAL_NUM to single, double top
/**metal for 1P4M is not allowed!
#DEFINE TOP_METAL_NUM single
//=====
/**OPTION 4: Define ERC Check or Not. The value can be TRUE or FALSE(Upper Case).
/**Default value is TRUE.
#DEFINE ERCCHECK FALSE
//=====
/**OPTION 5: Define XRC RUN or not.The value can be TRUE or FALSE(Upper Case) .
/**Default value is FALSE.
#DEFINE XRC_RUN TRUE
```



[Batch mode] STEP 2: Include PEX Databases into LVS

- Include Calibre xRC rule files (rules.C, rules.R and rules.S) into LVS runset. Note that the included rule files of which the metallization options should be consistent to the settings in the previous step.

```
#DEFINE NWELL_NOT_TO_POWER_CHECK FALSE
//=====
/**OPTION 27: turn on pwell not connecting to ground check or not. The value can be TRUE or FALSE(Upper Case).
**Default value is FALSE.

#DEFINE PWELL_NOT_TO_GROUND_CHECK FALSE
//=====

INCLUDE "/netapp/userhome/e028161/TECH_FILE/40LL/calibrexc_40LL_1P7M_5Ic_2Tmc_ALPA14.5k_v1.4_1R/rulefiles/TYPICAL/rules.R"
INCLUDE "/netapp/userhome/e028161/TECH_FILE/40LL/calibrexc_40LL_1P7M_5Ic_2Tmc_ALPA14.5k_v1.4_1R/rulefiles/TYPICAL/rules.C"
INCLUDE "/netapp/userhome/e028161/TECH_FILE/40LL/calibrexc_40LL_1P7M_5Ic_2Tmc_ALPA14.5k_v1.4_1R/rulefiles/TYPICAL/rules.S"

//=====
//Device tolerance setting
//user can change below tolerance according to the design
//=====
VARIABLE logmos_err 5
VARIABLE dio_err 5
VARIABLE res_err 5
VARIABLE bjt_err 5
VARIABLE mom_err 5
VARIABLE var_err 5
VARIABLE rfmos_err 5
VARIABLE ind_err 5
VARIABLE ldmos_err 5
VARIABLE sram_err 5
//=====
```



[Batch mode] STEP 3: Define SOURCE and LAYOUT in LVS

- Follow the usage of Calibre LVS, specify the SOURCE, LAYOUT in the runset, respectively.

```
VARIABLE ldmos_err 5  
VARIABLE sram_err 5  
//=====
```

```
SOURCE PATH "INV.sp"  
SOURCE PRIMARY "INV"  
SOURCE SYSTEM SPICE
```

CDL / SPICE netlist

```
LAYOUT PATH "INV.gds"  
LAYOUT PRIMARY "INV"  
LAYOUT SYSTEM GDSII
```

GDS file

```
LAYOUT BASE CELL "mom_2t_ckt*"  
LAYOUT BASE CELL "mom_3t_ckt*"
```

```
PRECISION 1000
```

```
//LAYOUT CASE YES
```



[Batch mode] STEP 4: Define Output Netlist Format

- Specify the output PEX netlist format in the LVS.

```
Terminal
File Edit View Terminal Tabs Help
//LAYOUT CASE YES
//SOURCE CASE YES

LVS REPORT "lvs.rep"
LVS REPORT OPTION A B C D S

UNIT CAPACITANCE FF
UNIT RESISTANCE OHM
UNIT LENGTH U

#TEDEF XRC RUN TRUE
PEX NETLIST LUMPED "lumped.dist" HSPICE LAYOUT GROUND VSS MASK
PEX REPORT LUMPED none
PEX REPORT DISTRIBUTED none
PEX NETLIST DISTRIBUTED "distributed.dist" HSPICE LAYOUT GROUND VSS MASK
MASK SVDB DIRECTORY "svdb" XRC QUERY
#ELSE
MASK SVDB DIRECTORY "svdb" QUERY
#ENDIF

ERC MAXIMUM RESULTS 100
ERC RESULTS DATABASE erc.db
ERC SUMMARY REPORT erc.rep
```



[Batch mode] Step 5: Run Calibre xRC by Batch Mode

- In order to prevent the parasitic R, C parameters of RF devices to be double extracted, SMIC provide hierarchical and hybrid flow by applying *hcell_list* and *xcell_list*. The *hcell_list* and *xcell_list* in XRC package are example. User should add device base on his design.
- It requires 3 stages for parasitic extractions as follows,

- Stage 1: Command Syntax for Calibre LVS-H

```
calibre -lvs -hier -hcell hcell_list -spice ./svdb/<top_cell>.sp  
<Calibre_LVS_runset>
```

- Stage 2: Extracting Parasitics

```
calibre -xrc -pdb -rcc -xcell xcell_list <Calibre_LVS_runset> | tee  
1.log
```

- Stage 3: Generating an Extracted Netlist

```
calibre -xrc -fmt -all <Calibre_LVS_runset> | tee 2.log
```

where <top_cell> is the top cell name of the design

<Calibre_LVS_runset> is the Calibre LVS runsets provided by SMIC

Note: the *red* part could be modified based on your preferring extraction mode.



[Batch mode] Step 6: Extraction Results

- Three files will be created after running Calibre XRC.
- Please include *.pex.netlist to run post-layout simulation.

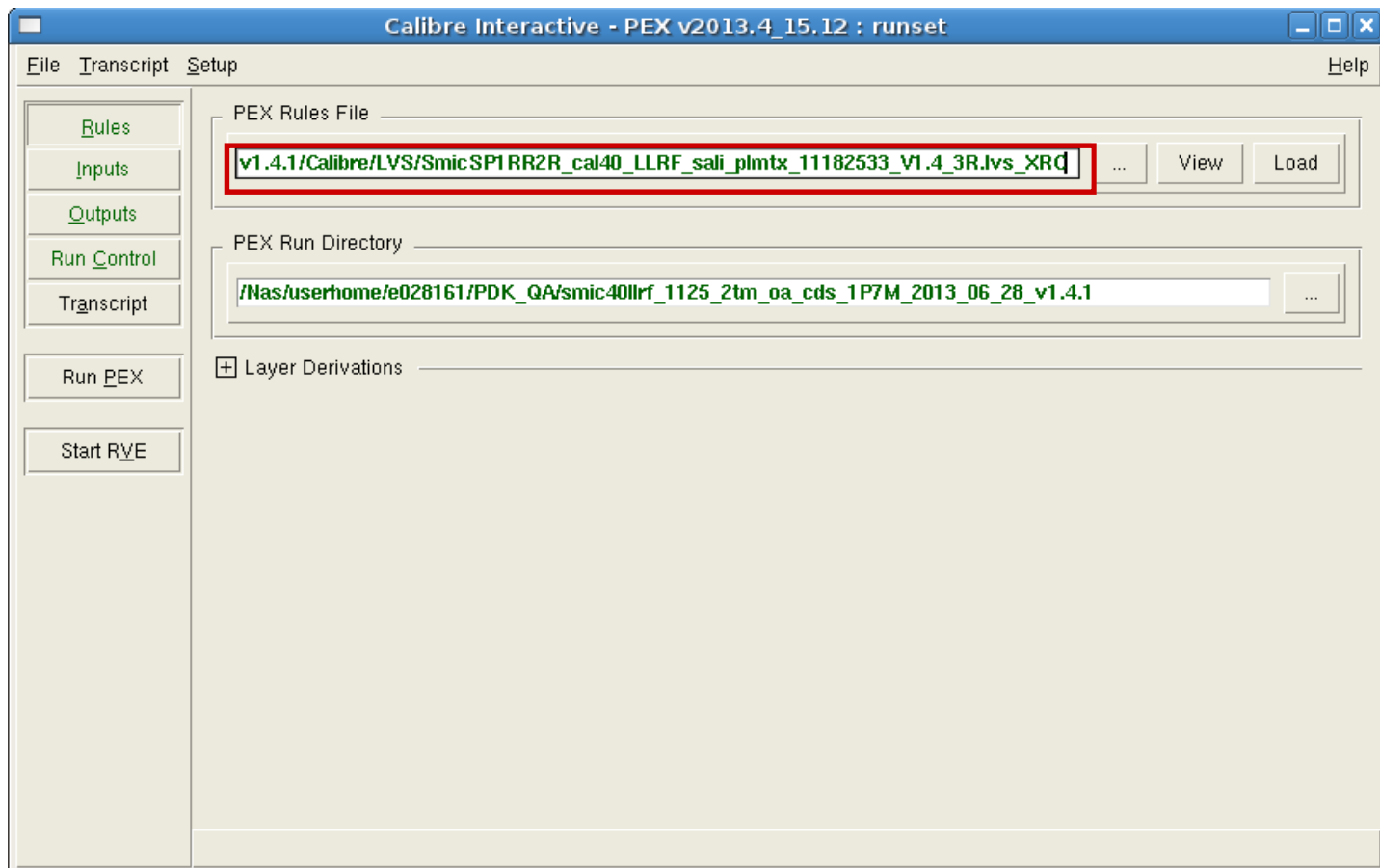
INV.pex.netlist
INV.pex.netlist.pex
INV.pex.netlist.INV.pxi

```
PEX Netlist File - INV.pex.netlist
File Edit Options Windows
* File: INV.pex.netlist
* Created: Tue Jul 28 16:37:55 2015
* Program "Calibre xRC"
* Version "v2013.4_15.12"
*
.include "INV.pex.netlist.pex"
.subckt INV OUT VDD! VSS! IN
*
* IN IN
* VSS! VSS!
* VDD! VDD!
* OUT OUT
DDO N VDD! DDO_pos N VSS! DDO_neg PARASITIC_NWD AREA=1.2376e-12 PJ=5e-06 M=1
XNM0 N OUT XNM0_DRN N IN XNM0_GATE N VDD! XNM0_SRC N VDD! DDO_pos N11LL CKT
+ W=2e-06 L=5e-08 AS=2.4e-13 AD=2.4e-13 PS=4.24e-06 PD=4.24e-06 SA=1.2e-07
+ SB=1.2e-07 SCA=1.55944 SCB=0.000169395 SCC=3.7517e-07 NRS=0.06 NRD=0.06
+ DPS=1.5e-07 DSTS=5.75e-07 DPCS=5e-08 PRELAYOUT=0 DCN=0 LPEMOD=1 MISMOD=1
+ GLOBALMOD=1 NF=1 MR=1
XPM0 N OUT XPM0_DRN N IN XPM0_GATE N VSS! XPM0_SRC N VSS! DDO_neg P11LL CKT
+ W=1.5e-06 L=5e-08 AS=1.8e-13 AD=1.8e-13 PS=3.24e-06 PD=3.24e-06 SA=1.2e-07
+ SB=1.2e-07 SCA=39.9692 SCB=0.0421454 SCC=0.00451151 NRS=0.08 NRD=0.08
+ DPS=1.5e-07 DSTS=5.75e-07 DPCS=5e-08 PRELAYOUT=0 DCN=0 LPEMOD=1 MISMOD=1
+ GLOBALMOD=1 NF=1 MR=1
x3 PM_INV*OUT_X3_1 PM_INV*VSS!_X3_2 MOM_2T_CKT_CDNS_437985212602
*
.include "INV.pex.netlist.INV.pxi"
*
.ends
*
*
.subckt MOM_2T_CKT_CDNS_437985212602 1 2
X0_noxref 1 2 MOM_2T_CKT L=8e-06 NN=40 TM=4 BM=2 MISMOD=1 FLAG_WIDTH=0 MM=1
.ends
*
*
```



[GUI mode] Set Calibre Interactive (I)

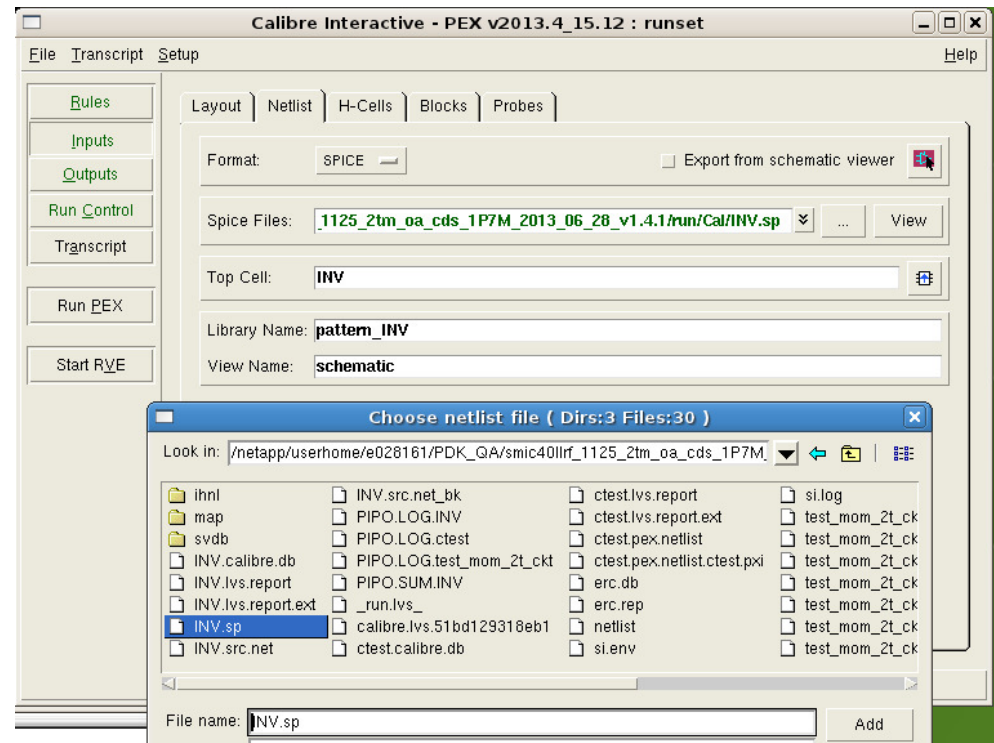
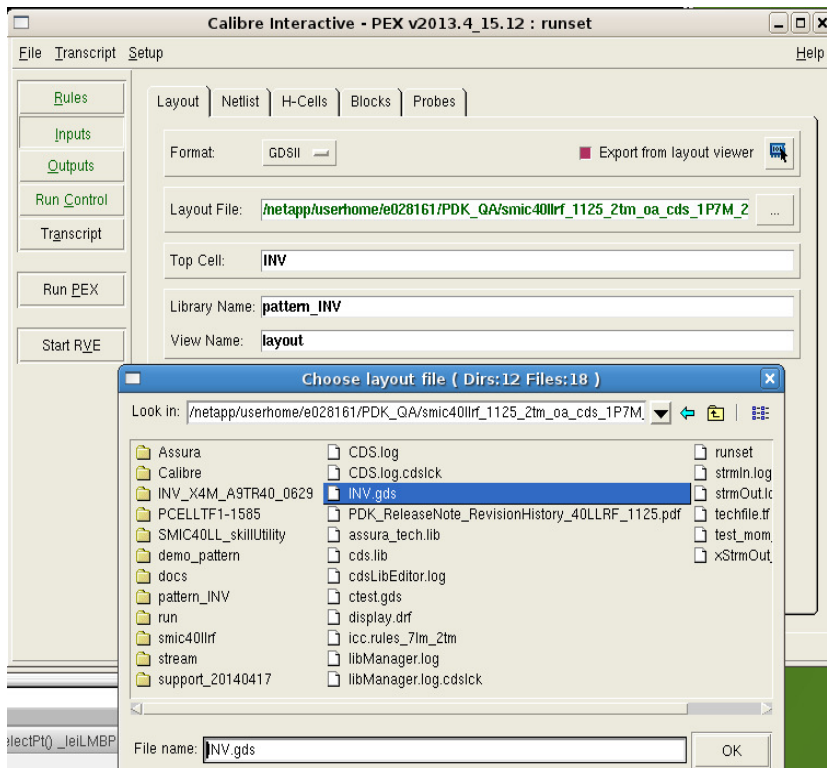
- In the *Layout Editing* window, user could select *Calibre* → *Run PEX* to invoke Calibre Interactive as follows. Browse to select your revised LVS file in the *PEX Rules File* column.





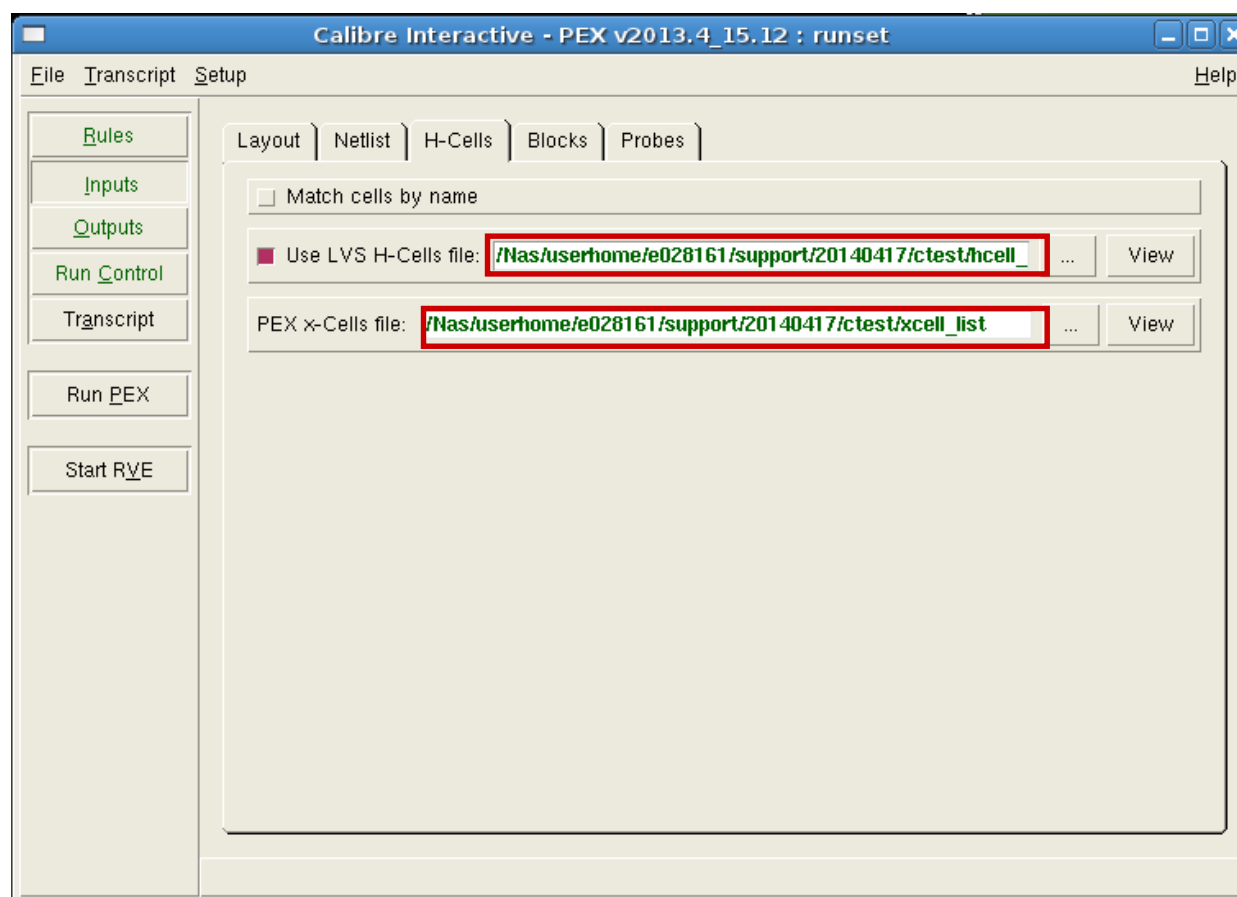
[GUI mode] Set Calibre Interactive (II)

- Switch to the **Input** section. Browse to select the design GDS in the **Layout** sheet and the design CDL netlist in the **Netlist** sheet, respectively.



[GUI mode] Set Calibre Interactive (III)

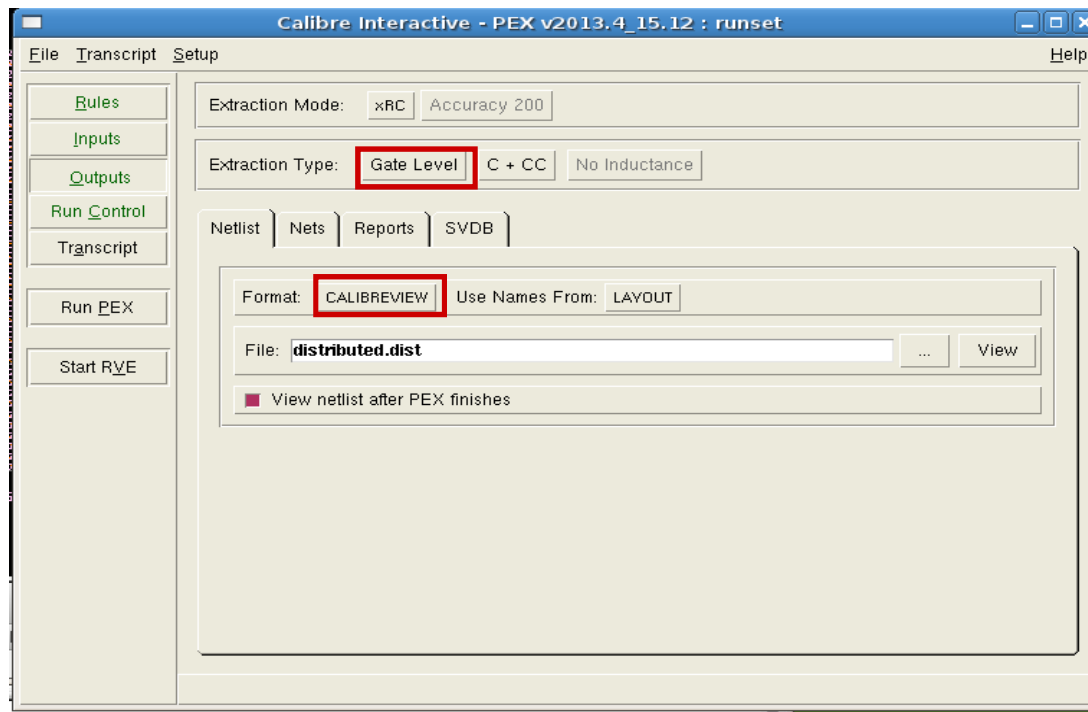
- Familiar to the settings of the batch mode, we also need to include *hcell_list* and *xcell_list* to invoke the hierarchical and hybrid extraction flow.





[GUI mode] Set Calibre Interactive (IV)

- Switch to the **Output** section. In this section, user could decide the **Extraction Mode**, **Extraction Type** and **Output Netlist Format** according to their demand.
- For hierarchical and hybrid flow, **Gate Level** should be chosen. Otherwise, please choose **Transistor Level** as shown in right side.
- Here we select **CALIBREVIEW** to continue the following demonstration.



[GUI mode] Run PEX and Generate Calibre View (I)

- I, Usually, Schematic library and layout library will be generated in the same topcell, but when you separate schematic library and layout library as individual library, please fill your schematic library name in this item.

- II, Now we could click **Run PEX** to start the parasitic extraction. When the job is concluded, **Calibre View Setup** window will pop-up. Browse to select **Cellmap File**, and then click OK.

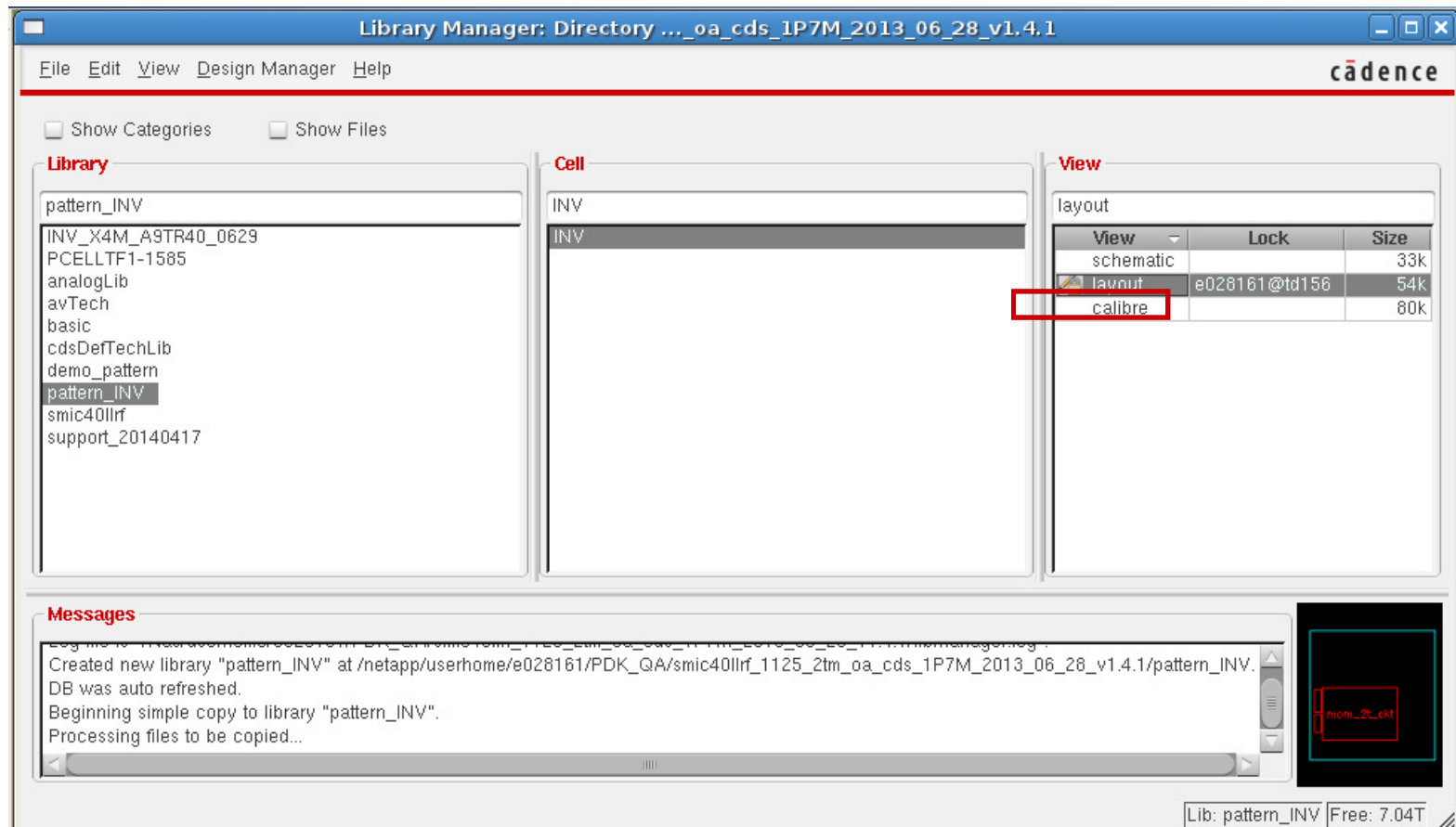
The screenshot shows the 'Calibre View Setup' dialog box with the following fields and options:

- CalibreView Setup File: [Empty field] (Buttons: Browse, View, Load)
- CalibreView Netlist File: ./2tm_oa_cds_1P7M_2013_06_28_v1.4.1/run/Cal/INV.pex.netlist (Button: Browse)
- Output Library: pattern_INV
- Schematic Library: pattern_INV (Highlighted with a red box and a red arrow from the text 'fill your schematic library name in this item')
- Cellmap File: ./calview.cellmap (Highlighted with a red box and a red arrow from the text 'Browse to select Cellmap File')
- Log File: ./calview.log (Buttons: View, Edit, Browse)
- Calibre View Name: calibre
- Calibre View Type: ☒ maskLayout ☐ schematic
- Create Terminals: ☒ if matching terminal exists on symbol ☐ Create all terminals
- Preserve Device Case: ☐
- Execute Callbacks: ☐
- Suppress Notes: ☐
- Reset Properties: m=1
- Magnify Instances By: 1
- Device Placement: ☒ Layout Location ☐ Arrayed
- Parasitic Placement: ☐ Layout Location ☒ Arrayed
- Buttons: OK, Cancel, Save, Help



[GUI mode] Run PEX and Generate Calibre View (II)

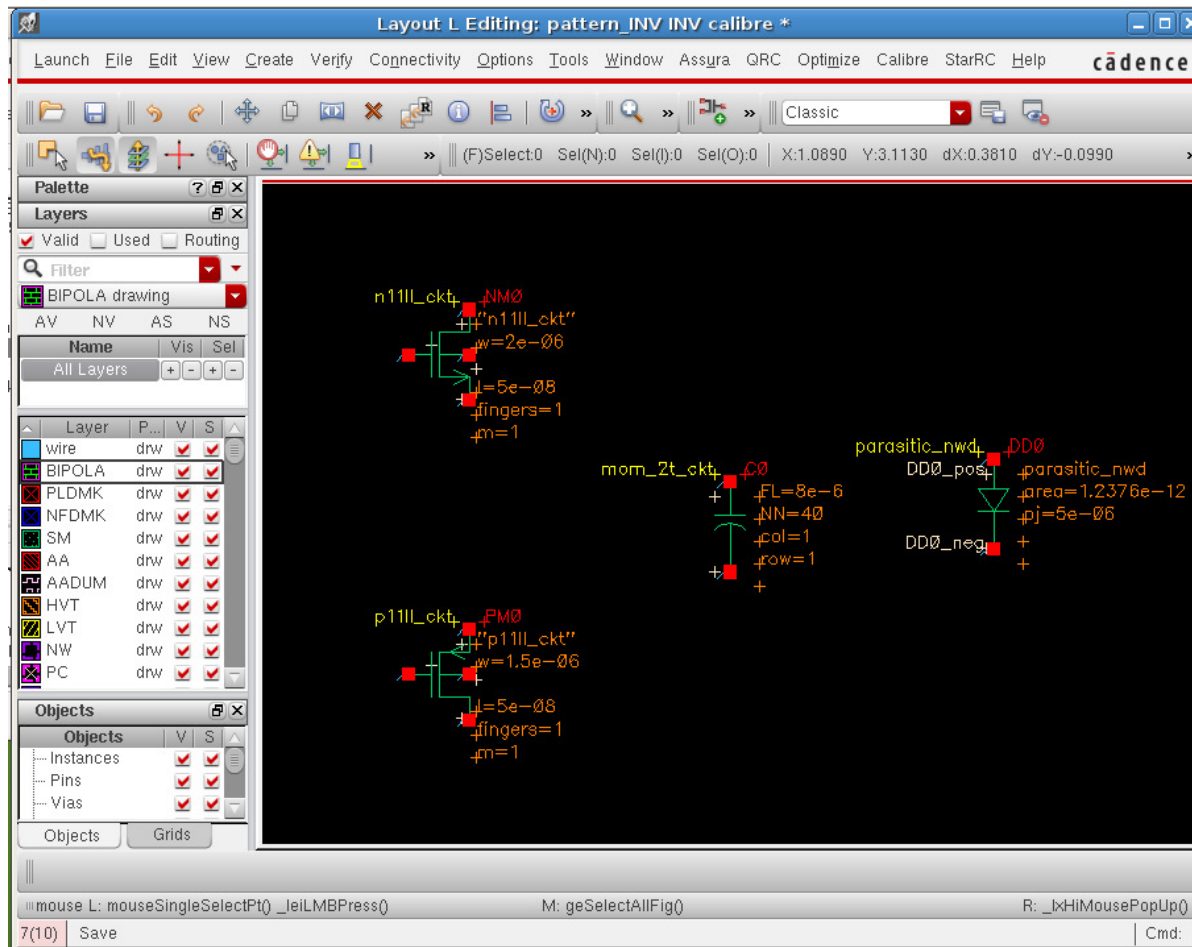
- Now we could find *calibre* view of the cell *INV* appears in the *Library Manager* window.





[GUI mode] Extraction Results (I)

- All devices are correctly extracted and annotated in Calibre View, matching to the geometry in layout.



Post-layout Extraction Team



[GUI mode] Extraction Results (II)

- A large amount of the parasitic RC components are extracted and annotated in Calibre View. Their influence will be reflected in post-layout simulation.

